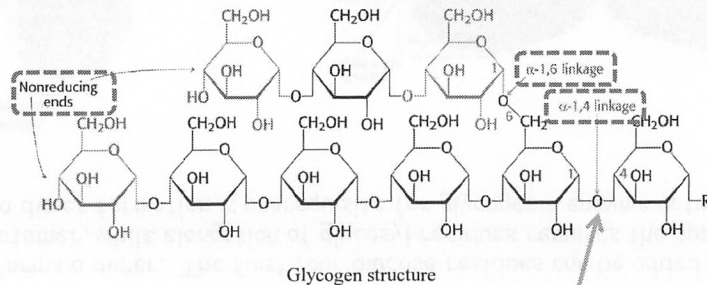


Glycogen Metabolism

- Glycogen is a readily mobilized form of glucose.
- It is stored exclusively in liver and skeletal muscle.
- In a fed state, liver glycogen content accounts for as much as 10% of this organ, while skeletal muscle (♂ 34%, ♀ 27% BW) stores less glycogen (only 1-2%).
- However, total glycogen contents is twice larger in skeletal muscle (~80% of whole-body glycogen) than liver.



Glycogen is composed of glycosyl residues, most linked together by α -1,4-glycosidic linkages

Why store glucose as glycogen?

Why not fat?

1. Fat can not be mobilized nearly as rapidly as glycogen.
2. Fat can not be used as an energy source in the absence of oxygen.
3. Fat can not be converted to glucose to maintain blood glucose levels required by brain.

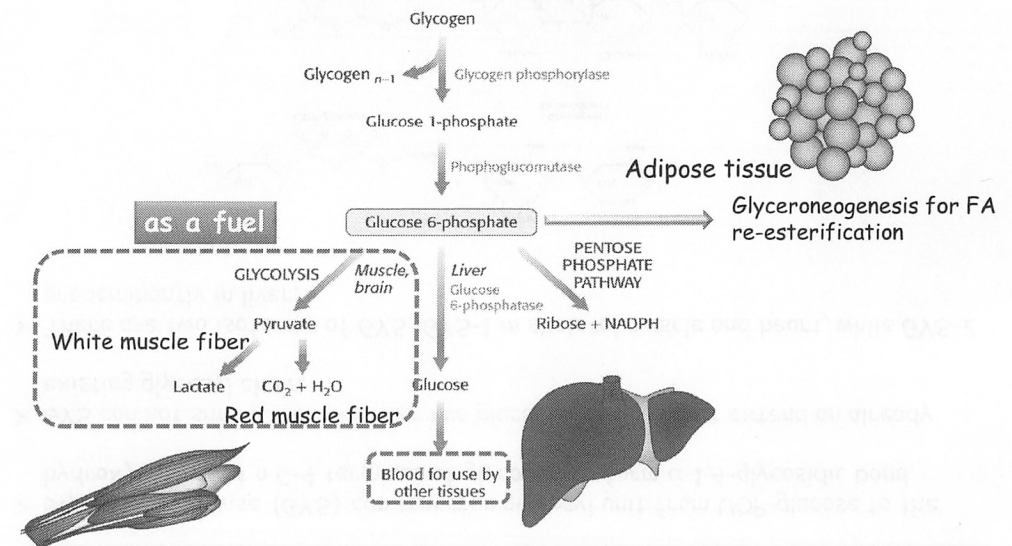
Why not glucose?

Glucose is osmotically active substance.

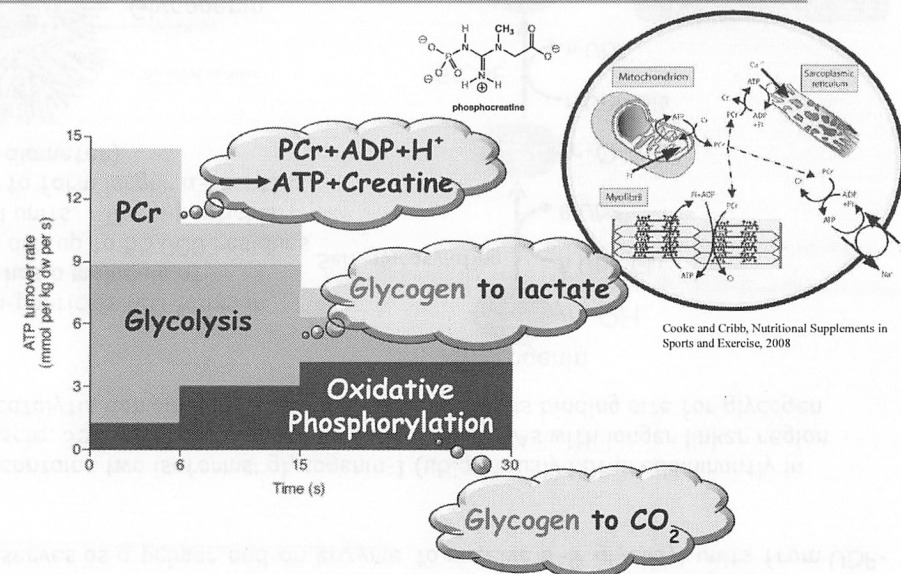
Concentration of glucose reaches about 400mM in hepatocytes to match the "glucose reserves" provided by hepatic glycogen contents.

Glycogen Metabolism

Physiological roles of glycogen stored are quite different between liver and skeletal muscles.



Glycogen is a major source for ATP production during short-term intense exercise such as throws, and jump and 100-m sprint



Anaerobic glycolytic capacity is approximately threefold higher than PCr

Glycogen synthesis is catalyzed by three enzymes

1. UDP-glucose pyrophosphorylase (G1P uridylyltransferase)
2. Glycogen synthase (GYS, rate-limiting step)
3. Branching enzyme

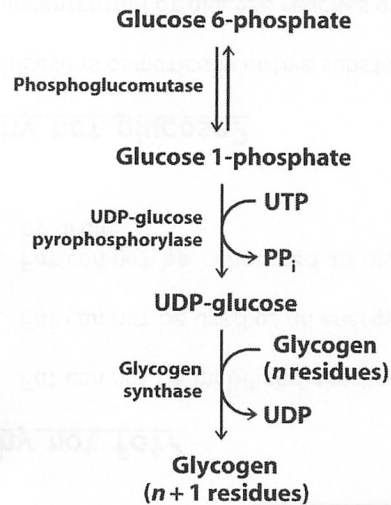
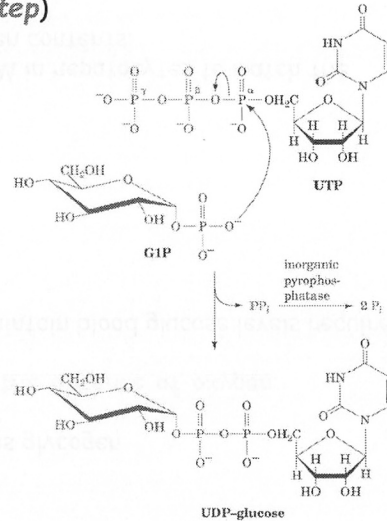


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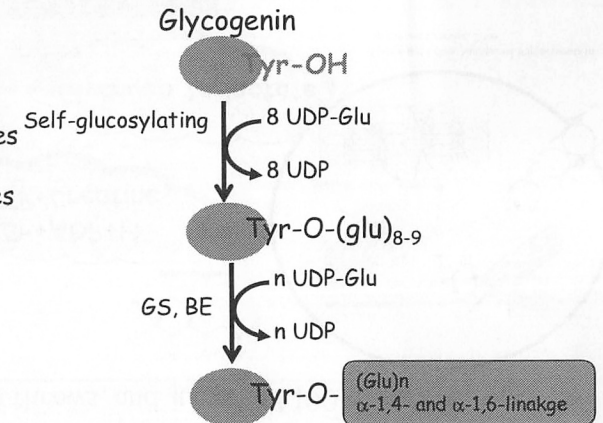
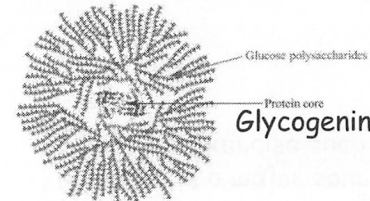
UDP-glucose serves as an activated form of glucose



Glycogenin is a core protein in glycogen particle

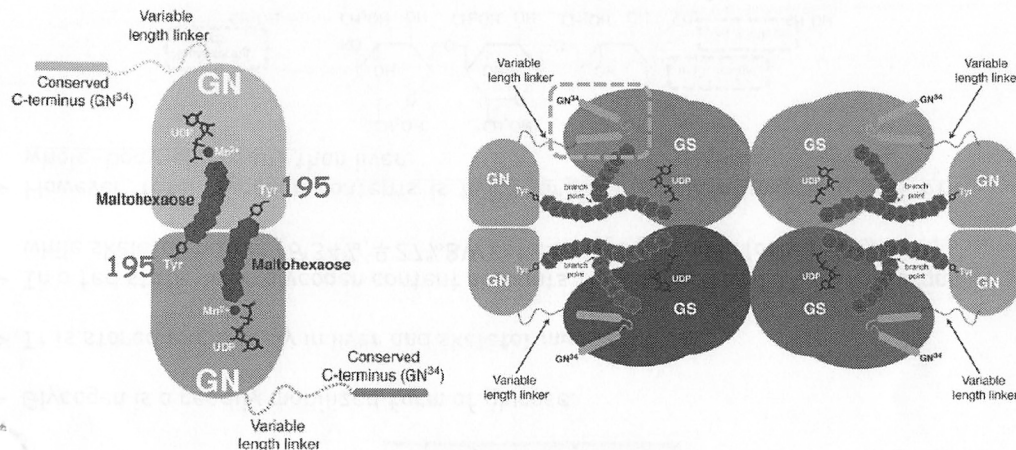
- Glycogenin serves as a primer and an enzyme to receive 8-9 glycosyl units from UDP-glucose.
- Glycogenin contains two isoforms: glycogenin-1 (ubiquitously but predominantly in skeletal muscle, 350 AAs) and glycogenin-2 (liver, 501 AAs with longer linker region between a catalytic domain and C-terminal domain that is binding site for glycogen synthase).

- Glycogen β-particles (20-50nm in diameter) has a molecule of glycogenin and up to 55,000 residues of glycosyl units. The β-particles aggregate to form larger α-rosettes (200nm in diameter).



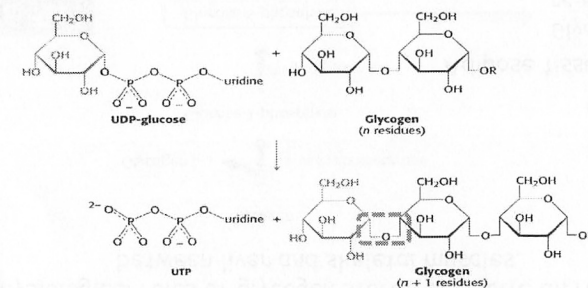
Glycogenin forms a dimer and binds to glycogen synthase via its C-terminal

- Glycogenin forms a dimer. The first four glucose residues can be added by the same catalytic protomer, while elongation of glycosyl residues requires the opposing protomer. Therefore, a dimer formation is prerequisite for glycogenin enzyme activity.



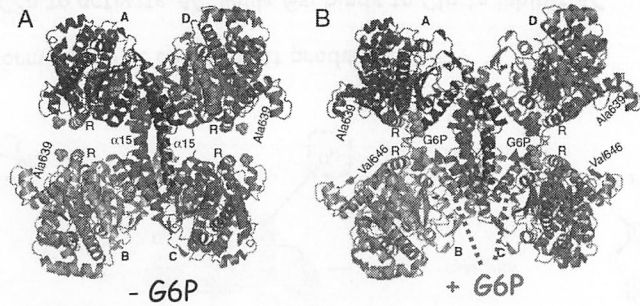
Glycogen synthase is the key regulatory enzyme in glycogen synthesis

- Glycogen synthase (GYS) can transfer glycosyl unit from UDP-glucose to the hydroxyl group at a C-4 terminus of glycogen to form α-1,4-glycosidic bond.
- GYS can not simply link together two glucose residues, but extend an already existing glycosyl chain.
- There are two isoforms of GYS; GYS-1 in skeletal muscle and heart, while GYS-2 predominantly in liver.



K_m of GS for glycogen (n) gets smaller as n increases

Glycogen synthase forms a tetramer which increases activity by binding to G6P



- GYS activity is **inhibited by phosphorylation** and allosterically **activated by G6P**.

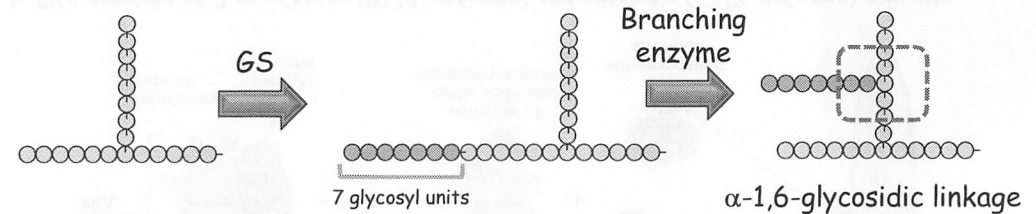
In liver, glucokinase can provide higher G6P with GYS to induce a conformational change and subsequently higher binding affinity to its substrate.

Branching enzyme is a bifunctional enzyme

When a straight-chain of 11 residues has been formed,

branching enzymes help **increase the solubility** of the glycogen particles in cytosol

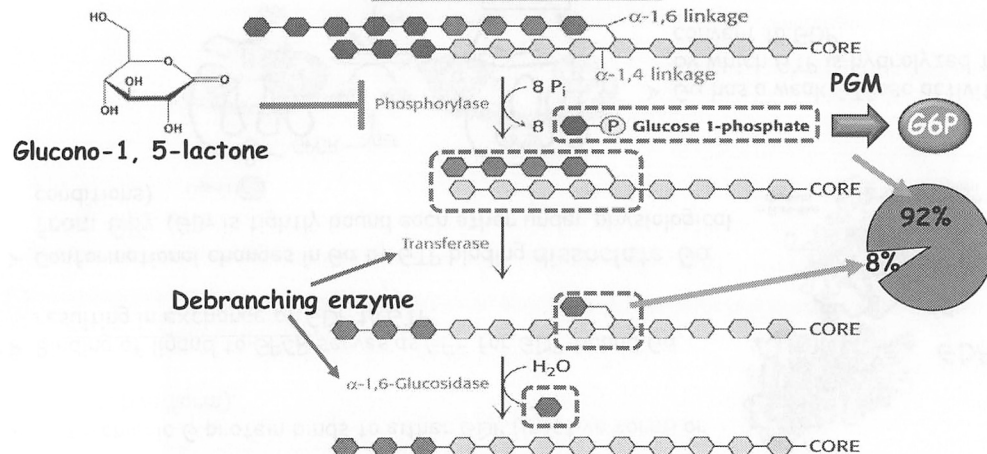
by transferring a block of residues, a typically 7 in number, to a more interior site on the same or another glycogen chain and form α -1,6-linkage.



Branching enzyme produces a large number of non-reducing ends of glucose.

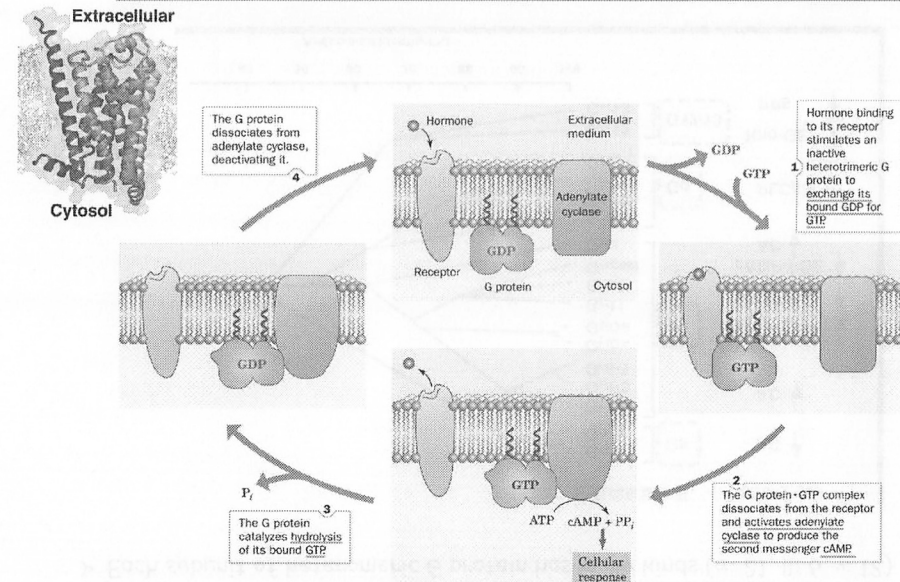
This allows efficient synthesis and breakdown of glycogen by providing many enzyme accessing sites.

Glycogen Breakdown



- Glycogen Phosphorylase breaks α -1,4-glycosidic linkage to produce **G1P**, but not glucose, until it reaches a terminal residue 4 glycosyl residues away from an α -1,6-branch point.
- Debranching enzyme has two activity; 4- α -D-glucanotransferase and amylo- α -1,6-glucosidase.
- An average molecule of glycogen yields about 12 molecules of G-1-P cleaved by glycogen phosphorylase for every molecule of glucose produced by debranching enzyme.

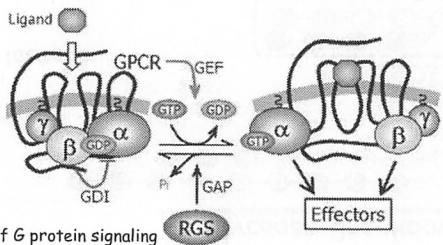
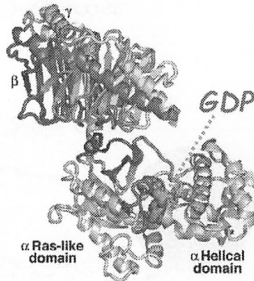
G protein-coupled receptor (GPCR)-mediated cell signaling



GPCR has 7 transmembrane helix domain and receives extracellular stimulus, leading to its conformational changes to transmit the signals into cells.

Heterotrimeric GTP-binding protein (Heterotrimeric G Protein)

- Heteromeric G protein is a member of regulatory GTPase superfamily, which also includes small G proteins involved in signal transduction (e.g. Ras), vesicular transport, actin filament formation and translation.
- Heteromeric G protein consists of three distinct subunits, α , β and γ subunit.
- Heteromeric G protein binds to either GDP (inactive form) or GTP (active form).
- Binding of ligand to GPCR serves as GEF for GDP-bound $G\alpha$, resulting in exchange of GDP to GTP.
- Conformational changes in $G\alpha$ by GTP binding dissociate $G\alpha$ from $G\beta\gamma$. ($G\beta\gamma$ is tightly bound each other under physiological conditions)

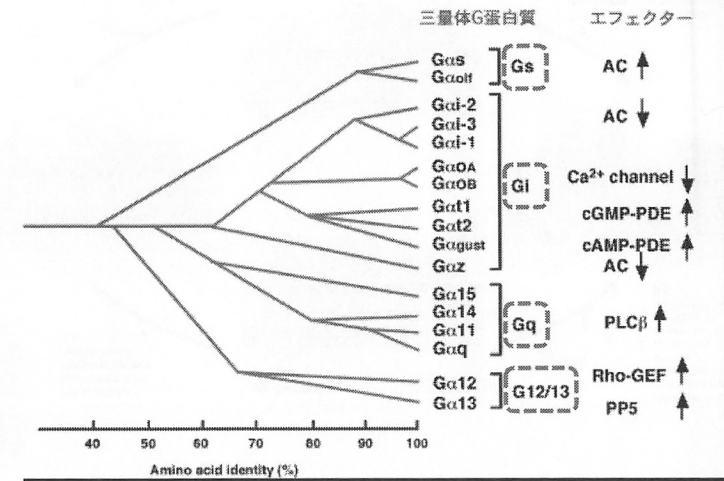


$G\alpha$ has a weak GTPase activity by which GTP is hydrolyzed to convert to GDP.

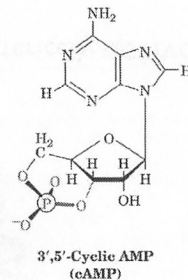
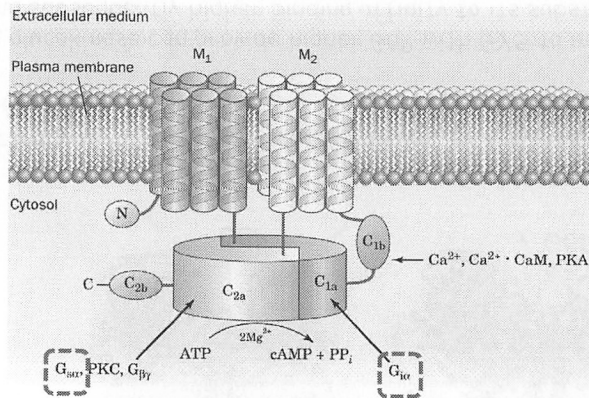
RGS: Regulator of G protein signaling

There are 4 categories among $G\alpha$ subunit by its targets

- Each subunit of heteromeric G protein has many kinds (α : 21, β : 6, γ : 12)

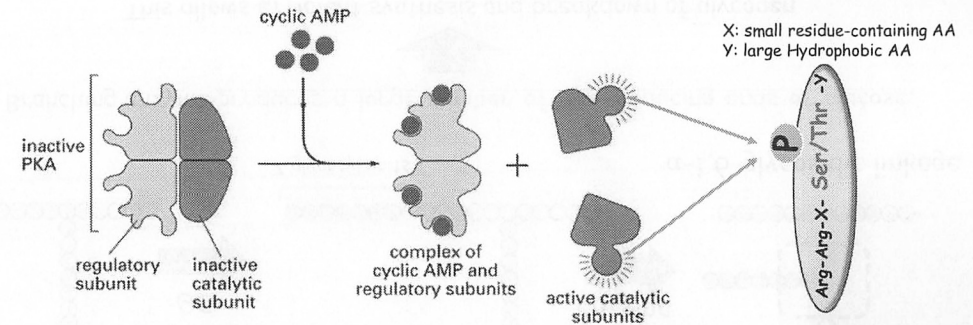


Adenylate cyclase (AC) generates a second messenger, cyclic AMP (cAMP)



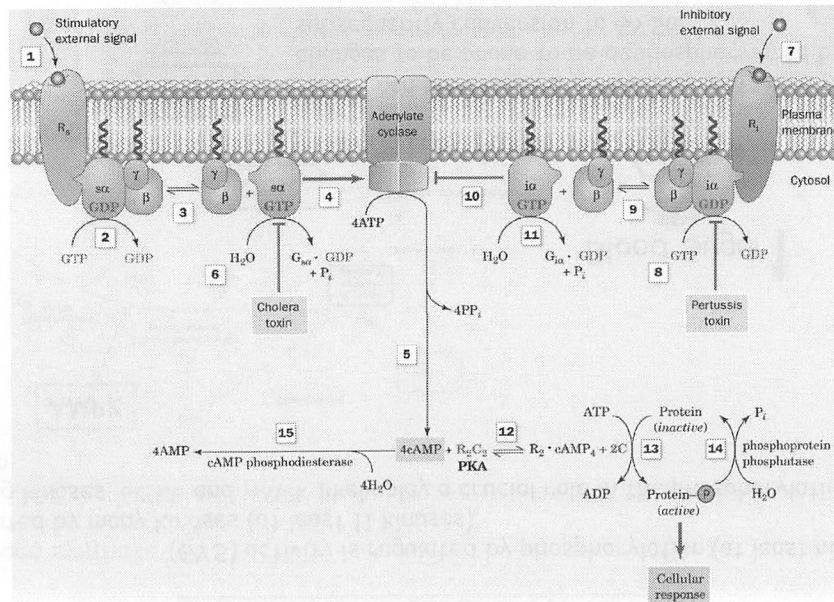
- C1a and C2a form catalytic center that produces cAMP.
- $G\alpha_s$ binds to C2a to activate AC, while $G\alpha_i$ binds to C1a to inhibit AC.
- Cholera toxin and Pertussis toxin induces ADP ribosylation of $G\alpha_s$ and $G\alpha_i$, respectively, resulting in constitutive activation and inhibition of its respective subunit (2枚後参照).

Cyclic AMP is a second messenger to activate protein kinase A (PKA)



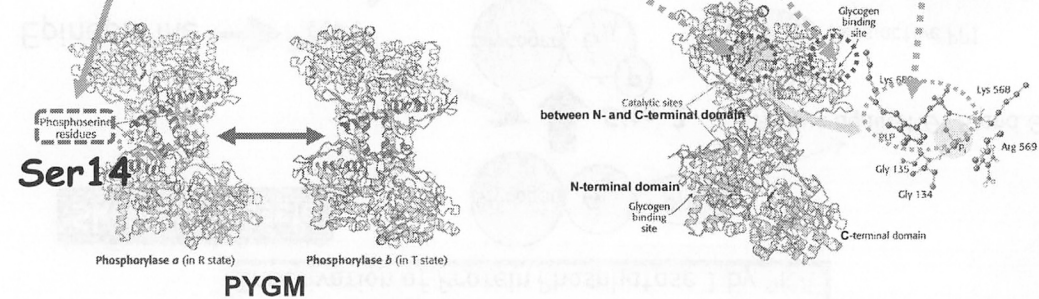
- PKA consists of 2 regulatory (R) (4 isoforms) and catalytic (C) (3 isoforms) subunits.
- When cAMP binds to R subunits, C subunits dissociate and are activated to phosphorylate Ser/Thr residue of its target proteins.
- A kinase-anchoring proteins (AKAPs) bind to R subunits of PKA by which PKA is targeted to different subcellular localization. This binding allows for local cAMP-responsive events to occur within specific compartment of the cells. In addition, AKAPs also show high affinities to other classes of signaling molecules such as PKC, PP1 and PP2B, through which AKAPs complexes allow cells to efficiently and accurately respond to signals.

GPCR-Adenylate cyclase-cAMP-PKA signaling pathway



Glycogen Phosphorylase

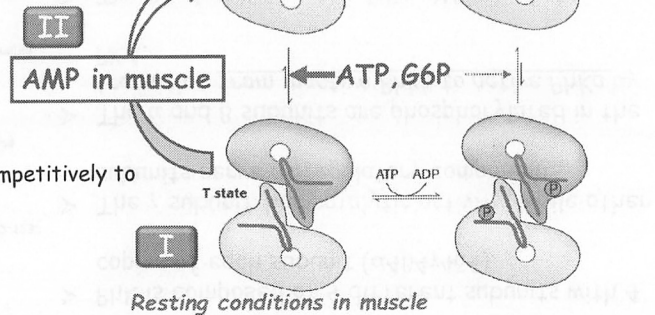
- There are three isoforms: liver (PYGL), muscle (PYGM), and brain (PYGB). They are highly homologues with distinct sensitivity to allosteric regulation and phosphorylation. PYGB is highly expressed in fetus.
- Glycogen phosphorylase (GP) is a homodimeric enzyme and requires pyridoxal 5'-phosphate (PLP), a derivative of Vitamin B6, as a co-factor, bound to Lys360 inside in catalytic site.
- GP exists in an active a form and an inactive b form which are interconverted by phosphorylase kinase (a sole kinase for GP)
- GP has five regulatory sites: catalytic site (glucose, G1P, Pi), glycogen site, AMP site (AMP, ATP, G6P), phosphorylation site, and indole site.



Allosteric and covalent-modified regulation of glycogen phosphorylase in skeletal muscle

Skeletal Muscle

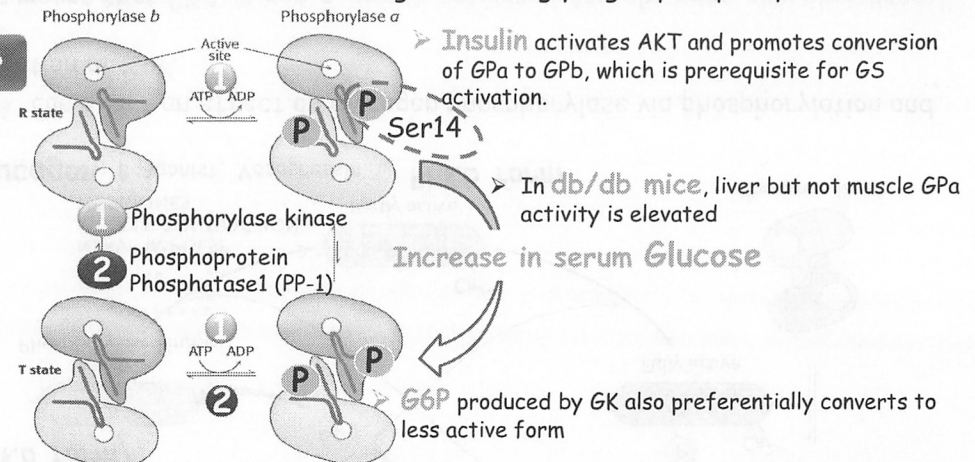
- AMP activates preferentially de-phosphorylated muscle, but not liver type of GP (See scheme II)



- Nearly all the enzyme of resting muscle is in inactive GPb (T state).
- When exercise commences, elevated AMP binds to GPb (T form) and converts to GPb (R state). AMP-bound GPb shows 80% of the activity of GPa, supporting glycogenolysis during exercise.
- Exercise also releases epinephrine which activates PKA (via GP kinase, but not GP directly!!) to phosphorylate GPb to GPa (full activity).

Allosteric and covalent-modified regulation of glycogen phosphorylase in liver

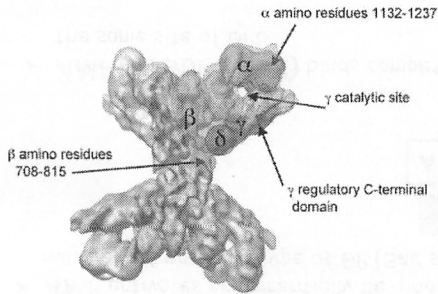
Liver



- GPa, but not GPb, exhibits the most responsive T-to-R transition by glucose (binding to catalytic site).
- Glucose-bound GPa is a better substrate for PP1, which converts GPa to GPb.
- AMP can not activate liver GPb as seen in skeletal muscle.
- Unlike muscle type, liver GPb lacks its catalytic activity, suggesting that phosphorylation of GP determines glycogenolysis

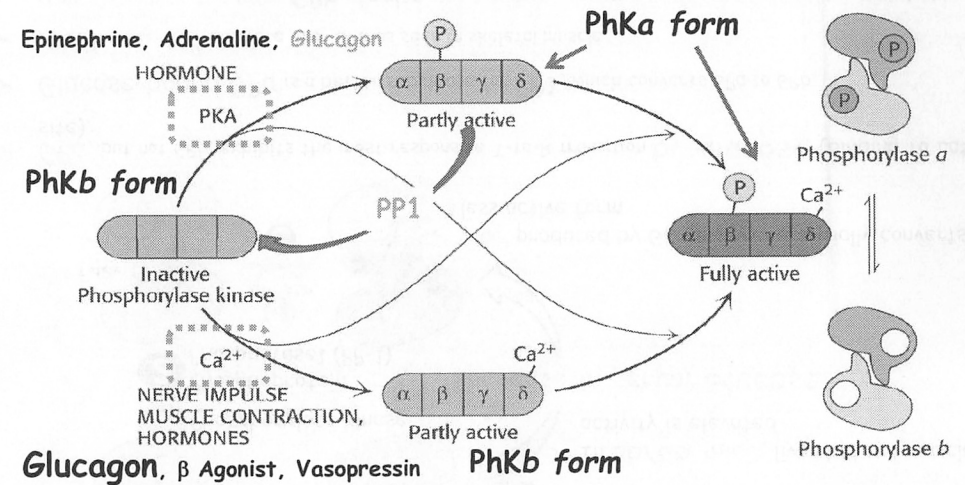
Activation of phosphorylase kinase (PhK)

Glycogen phosphorylase activity is regulated by phosphorylation at one site (Ser14) by one specific kinase, **phosphorylase kinase (PhK)**.



- PhK is composed of 4 different subunits with 4 copies of each subunit ($\alpha_4\beta_4\gamma_4\delta_4$).
- The γ subunit has catalytic activity, while other subunits serve as regulatory components.
- The α and β subunits are phosphorylated in the transition from inactive Phkb to active PhKa by **PKA**.
- The δ subunit is calmodulin. When subtle increase in intracellular Ca^{2+} levels occurs, Ca^{2+} -bound calmodulin induces conformational changes of PhK to be activated.

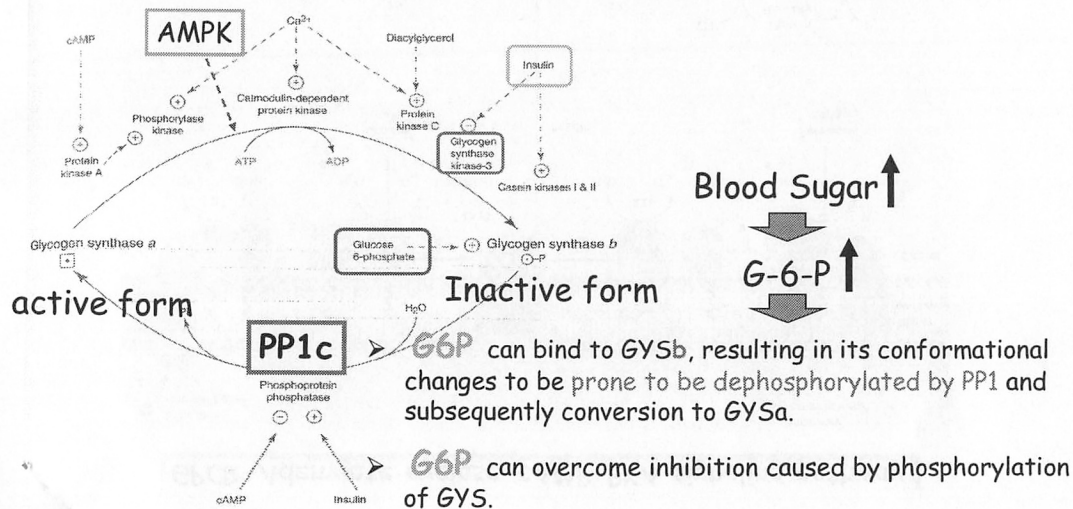
Activation of phosphorylase kinase (PhK)



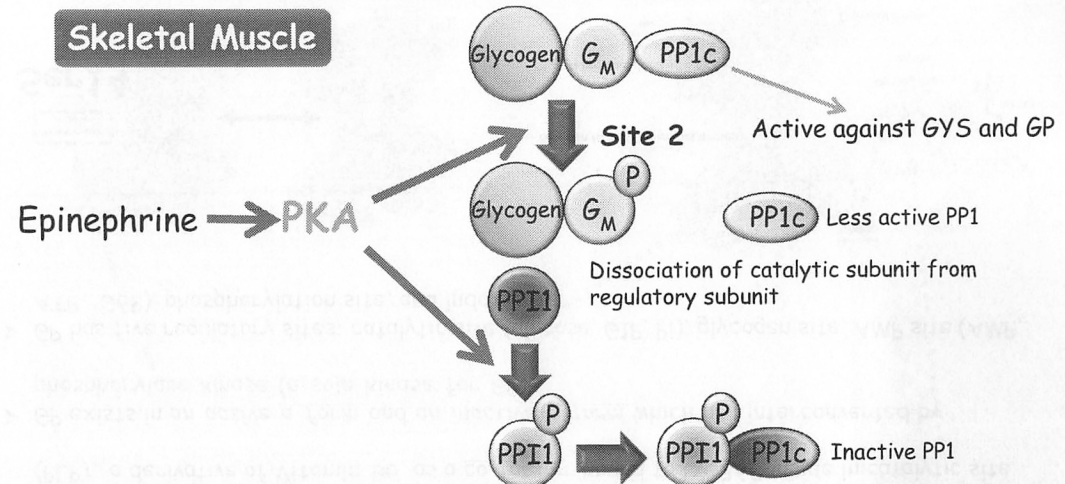
- **PKA** can exert an effect on glycogen phosphorylase via phosphorylation and activation of **PhK**.
- This means that PKA is not a direct activator for glycogen phosphorylase !!

Regulation of Glycogen Synthase

- Glycogen synthase (GYS) activity is regulated by phosphorylation (at least nine sites) mediated by many kinases (at least 11 kinases).
- Among kinases, **GSK3** and **AMPK** likely play a crucial role in the phosphorylation of GYS in vivo



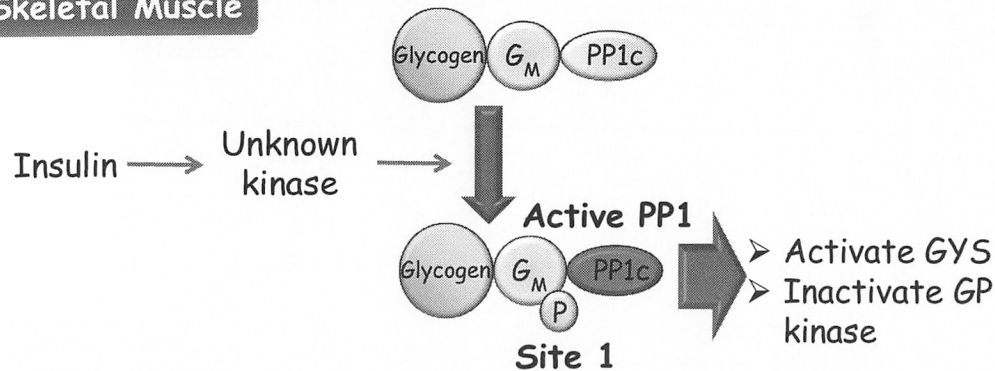
Inactivation of Protein Phosphatase 1 by PKA



- PP1 is **active** only when bound to glycogen via G_M subunit, a regulatory subunit of PP1 complex for glycogen metabolism.
- PKA phosphorylates **site 2** of G_M subunit, resulting in dissociation of catalytic subunit (PPI1) of PP1 complex.
- **Protein phosphatase Inhibitor 1**, another regulatory subunit of PP1 complex, inhibits cytosolic PP1 activity released from G_M subunit when phosphorylated by **PKA**.

Activation of Protein Phosphatase 1 by insulin

Skeletal Muscle



Insulin-PI3K signaling pathway phosphorylates site 1 of G_M subunit and forms a tight association between glycogen, G subunit and catalytic subunit (PP1c) of PP1 complex, resulting in activation of PP1.

悪性高熱症は、特に全身麻酔薬のハロセンなど、ある薬剤に曝露された場合に、著しい体温上昇、アシドーシス、高カリウム血症、筋硬直を来す遺伝的疾患である。この患者では、1型リアノジン受容体に変異があり、麻酔薬などによって、筋小胞体にある1型リアノジン受容体の変異体からカルシウムの異常な流出が起きる。これにより、解糖系やグリコーゲン代謝などの熱産生過程の制御が破綻する。

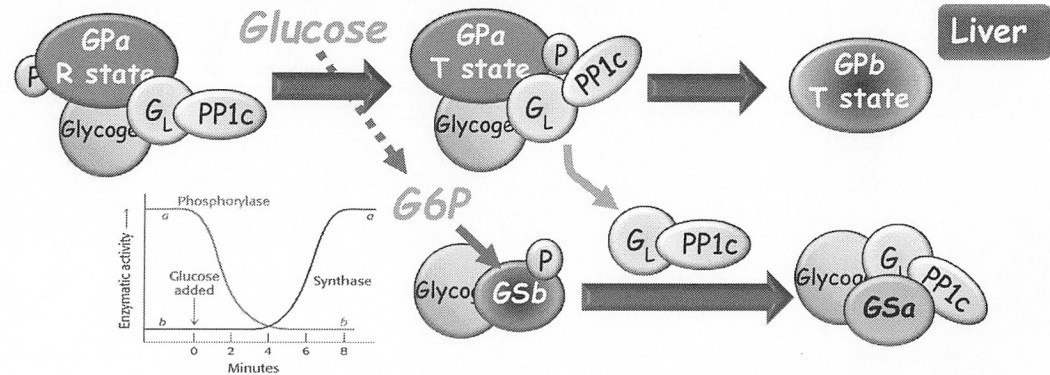
Q1: この患者では、なぜグリコーゲン分解が増加するのか。

Q2: グリコーゲン代謝制御において、生じないタンパク質修飾やアロステリック制御はどれか？

1. Phosphorylase kinaseのリン酸化
2. Phosphorylase bに対するAMPの結合
3. Phosphorylaseのリン酸化
4. Protein kinase Aのリン酸化
5. Glycogen synthaseの脱リン酸化

肝臓のグリコーゲンと骨格筋のグリコーゲンが、骨格筋のATP産生に利用される場合には、等量のグリコーゲンから産生されるATP量は同じか？

Glucose is a key regulator of glycogen metabolism in the liver



- G_L-PP1c complex tightly can bind to either R or T form of GPa (AMP site). However, PP1 can not access and dephosphorylate phosphor-Ser14 of GPa when serum glucose is low.
- When glucose levels increase, conformational change of GPa by glucose allows PP1 access to dephosphorylate GPa, resulting in conversion to GPb that loses affinity to G_L-PP1c complex.
- Released G_L-PP1c complex can access to other phosphorylated proteins such as GYSb, leading to conversion to active de-phosphorylated GYSa form. These events occur only when most GPa (over 90%) is converted to GPb by G_L-PP1c complex, because G_L-PP1c complex abundance is much lower compared to GP.
- G_L-PP1c complex is not regulated by phosphorylation like G_M-PP1c complex.

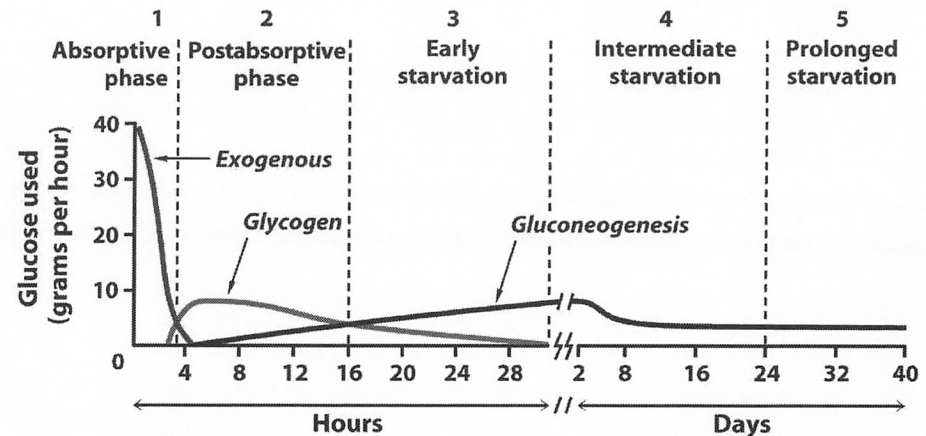


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